Revisiting the Inflammatory Pathway with Network Biology

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Introduction

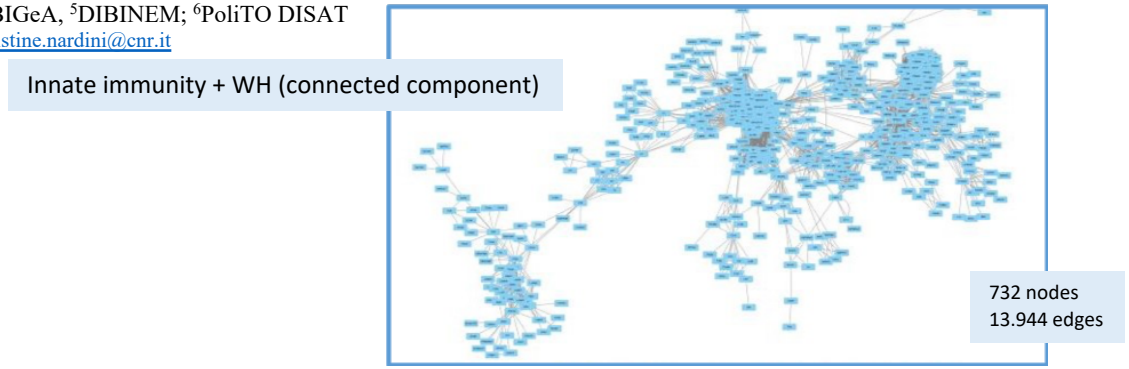
Inflammation is part of the complex function that addresses harmful stimuli and the first phase of *wound healing (WH)*, which guarantees living systems' homeostasis. Deviances from physiology make inflammation turn *acute* (sepsis, 11M death/y) or *chronic* (non-communicable diseases, 41M death/y). Therefore, tackling inflammation is a key priority. We recently proposed (Maturo, Maria Giovanna et al., 2020) to revise the *conventional* inflammatory pathway (innate immune response) to include *WH* (*expanded* inflammatory pathway).

Methods

We manually identified the Reactome pathways that include all reactions and species relevant to *WH*. Cytoscape was then used to perform the union of the SBML converted pathways, with the largest connected component being retained (732 nodes, 13.944 edges). The same was done for the *innate immune response* (R-HSA-168249.8) with 487 nodes, 11.744 edges. We then focused on: *NF-kB* (fundamental hub in all inflammatory reactions), *TNF-α* (renown target of inflammatory diseases) and *RAC1* (key player in mechano-transduction events of WH).

Results

Preliminary topological results highlight the stability of *closeness centrality*, i.e. all molecules preserve their efficiency in spreading information.



Conversely, *betweenness centrality* is stable for NF-kB (0.068), confirming NF-kB relevance, while halving its (very low) value in the *expanded* pathway for TNF-α (from 2.85E-06 to 1.29E-06). This indicates that the ability to bridge different parts of the graphs is less effective if we consider inflammation as an expanded concept, possibly contributing to explain the many side effects of anti-TNF-α therapies. Interestingly, RAC1 presents stable betweenness (from 0.094 to 0.093), comparable to NF-kB, supporting the hypothesis that WH leveraging therapies could act on a relevant and stable target, so far neglected (Nardini et al., 2016).

	Pathway	Betweenness Centrality	Closeness Centrality	Clustering Coefficient
NF-kB	WH+Innate Immunity	0.068	0.262	0.541
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TNF-α	WH+Innate Immunity	1.29E-06	0.209	0.833
	Innate Immunity	2.85E-06	0.242	0.833
RAC	WH+Innate Immunity	0.093	0.308	0.663
	Innate Immunity	0.094	0.371	0.757

Table 1: Table showing the preliminary network analysis performed